Phase-2 Detailed Research Report

Data-Backed Piecewise Milling Tool-Life Model, Validation, and Simulator Interpretation

Prepared from the local `phase -2` workspace

April 18, 2026

Abstract

This report documents the *phase-2* research effort contained in the local workspace under `phase-2/`. The objective of the work was not merely to fit a predictive curve, but to derive an interpretable and analytically invertible milling wear model that can serve as the core of a simulator. The evidence base combines three public datasets already present in the repository: the NUAA orthogonal milling bundle, the PHM 2010 cutter wear traces, and the NASA milling run-to-failure data. Rather than forcing these datasets into a single undifferentiated regression problem, the phase-2 strategy assigns each dataset a specific research role. NUAA supplies the parameter-dependent early wear law, PHM 2010 checks whether the early-regime exponent transfers to a second milling dataset, and NASA supplies the late-stage acceleration evidence that is absent from the shorter-life datasets.

The resulting model is a piecewise power law. The early regime follows a sub-linear exponent of 0.64; the late regime follows an accelerated exponent of 1.29. The fitted early amplitude law around the reference condition of 1800 rpm, 0.05 mm/tooth, and 3.0 mm axial depth is

$$\phi = \left(\frac{n}{1800}\right)^{0.3600} \left(\frac{f_z}{0.05}\right)^{3.9560} \left(\frac{a_p}{3.0}\right)^{0.6881},$$

with early wear amplitude coefficient $k_{\text{early}} = 0.018\,580\,9$. The final simulator preserves closed-form life prediction to a target wear threshold while exposing a calibration factor λ for tool-workpiece adaptation and a late-stage multiplier ρ for material-family effects.

This report is intentionally written from a research perspective. It records data provenance, preprocessing, model structure, fitted coefficients, bootstrap stability, comparison against machine-learning baselines, engineering interpretation, and limitations. It also adds native LaTeX diagrams to explain the data-role assignment, the piecewise wear law, and the calibration workflow. The purpose is to turn the phase-2 folder from a useful prototype into a manuscript-style technical artifact that another researcher or engineer can inspect, critique, and reproduce.

Contents

1 Introduction

The phase-2 workspace addresses a common challenge in prognostics and health management: useful public datasets exist, but no single dataset provides all of the information required to build a credible and interpretable end-to-end tool-life model. Some datasets are rich in process-parameter variation but terminate before catastrophic wear. Others continue into failure but operate over narrower process settings or different materials. As discussed more generally in PHM dataset reviews, practical research often proceeds under such partial and heterogeneous evidence conditions rather than under ideal run-to-failure coverage across the entire operating envelope [1].

The phase-2 codebase responds to that problem with a deliberate decomposition:

- use the NUAA orthogonal milling bundle to identify the *shape and parameter law* of early wear,
- use PHM 2010 wear traces to verify whether the early-regime exponent remains in the same range on a second milling dataset,
- use the NASA milling cases to quantify *late-stage acceleration* and material-family dependence.

This is a stronger research position than pretending that one dataset alone supports every coefficient in the final simulator. It acknowledges dataset limits directly and turns them into explicit modeling assumptions.

1.1 Research question

The central research question is:

> Can the local phase-2 evidence support an analytical milling wear law that is sufficiently interpretable, data-backed, and invertible to serve as a practical tool-life simulator?

That question contains four sub-problems:

1. Determine whether early milling wear is better represented by a power law than by a linear-in-time law.
2. Estimate how spindle speed, feed per tooth, and axial depth modulate early wear amplitude.
3. Determine whether late-stage wear accelerates, and whether that acceleration differs by material family.
4. Convert the final equation into a closed-form simulator that predicts wear trajectories and threshold life.

1.2 Artifacts reviewed

This report is grounded in direct inspection of the phase-2 implementation and outputs:

- `phase-2/research_pipeline.py`
- `phase-2/report_builder.py`
- `phase-2/tool_life_simulator.py`
- `phase-2/outputs/phase2_model_summary.json`
- `phase-2/outputs/*.csv`
- `phase-2/plots/*.png`
- `phase-2/sources.md`
- local dataset documentation in `data/uniwear/README.md` and `data/nasa_milling/Readme.pdf`

The document therefore reflects what the code *actually does*, not what one might loosely infer from the folder names.

1.3 Contributions of this report

Relative to the existing README and Word report, this LaTeX manuscript contributes:

- a more explicit research narrative that explains why the datasets were assigned different modeling roles,
- a full mathematical statement of the early, late, threshold-life, and calibration equations,
- a detailed interpretation of the validation outputs already generated in phase-2,
- native LaTeX diagrams for the evidence flow, regime structure, and calibration workflow,
- a more explicit discussion of limitations, assumptions, and threats to validity.

2 Evidence Base and Dataset Role Assignment

2.1 Dataset roles

The local workspace uses three public milling datasets whose value lies in *complementarity* rather than uniformity.

Table 1: Dataset roles in the phase-2 research design

Dataset	Primary role in phase-2	Series	Points	Why it matters
NUAA orthogonal bundle	Fit the early-regime exponent and parameter-dependent amplitude law	9	152	Supplies a clean orthogonal grid over spindle speed, feed per tooth, and axial depth. This is the only local source with direct variation across all three early-regime process inputs.
PHM 2010	Verify the early-regime exponent on a second milling dataset	3	945	Offers long cutter wear trajectories with dense cut-level wear labels, but in the local wear files does not expose the same parameter grid as NUAA. Therefore it is used to validate early wear shape, not the condition law.
NASA milling	Fit the late-regime acceleration exponent and estimate material-family late-stage ratios	16	146	Extends well into heavy flank wear and failure, with explicit material family, feed, and depth information. It is the key source for the accelerated post-transition regime.

The quantitative summary generated by `phase-2/outputs/dataset_profiles.csv` is consistent with those roles. NUAA spans 9 experiments and 152 run-level observations; PHM contributes 945 cut-level points across three cutters; NASA contributes 146 case-level points across 16 cases, including wear values up to 1.53 mm.

Table 2: Compact dataset coverage summary from the generated phase-2 outputs

Dataset	Series	Points	Minimum start wear (mm)	Maximum end wear (mm)	Maximum time/inde
NASA	16	146	0.0000	1.5300	105.0
NUAA	9	152	0.0600	0.3127	55.4
PHM2010	3	945	0.0242	0.2159	315.0

2.2 Why a multi-dataset regime design is justified

Trying to fit one universal law to all three datasets would obscure more than it would explain. The datasets differ in material systems, sensor instrumentation, time indexing, and depth of wear coverage. The more defensible move is to acknowledge these differences and ask a narrower question of each source.

That design choice mirrors a broader PHM reality: researchers often have partial lifecycle coverage, missing parameter sweeps, and heterogeneous measurement histories rather than one perfect run-to-failure archive [1]. The phase-2 model is therefore best understood as a *hybrid engineering law* assembled from complementary evidence:

1. early wear shape and parameter sensitivity from NUAA,
2. early wear consistency check from PHM 2010,
3. late wear acceleration and material dependence from NASA.

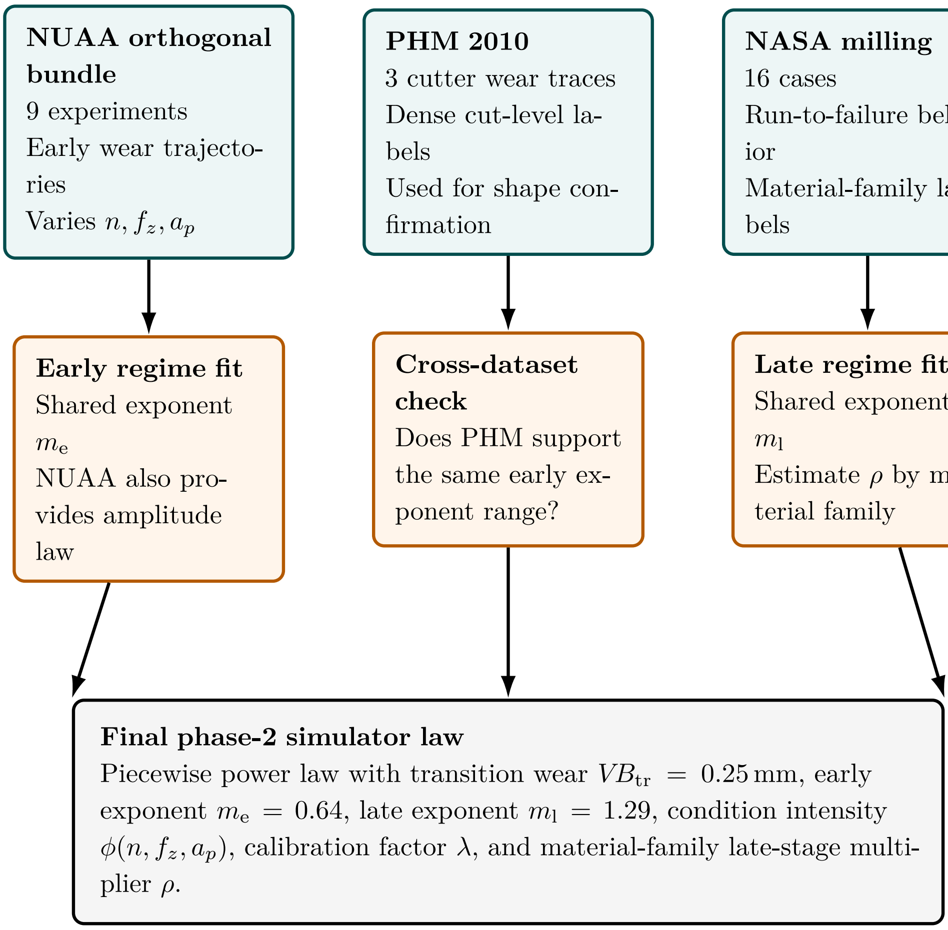

Figure 1: Native LaTeX diagram of the phase-2 evidence flow. Each dataset contributes to a specific modeling sub-problem rather than to one undifferentiated global fit.

2.3 Primary source notes

The local documentation supports the dataset-role assignment:

- The Uniwear README states that the NUAA orthogonal bundle contains the 3×3 combinations of feed per tooth $\{0.045, 0.05, 0.055\}$, spindle speed $\{1750, 1800, 1850\}$, and corresponding axial depth settings, making it the natural basis for a parametric early wear law [2].

- The same README notes that PHM 2010 contributes cutter wear traces and overlapping signal channels but, in the local phase-2 use case, the wear files are primarily exploited for wear labels rather than for an explicit parameter-law refit [2].
- The NASA documentation identifies the fixed cutting speed of 200 m/min (826 rpm), two depths of cut (0.75 mm, 1.5 mm), two feed values (0.25 mm/rev, 0.5 mm/rev), and two material families (cast iron and steel), making it the natural source for late-stage and material-family interpretation [3].

3 Data Preparation and Exploratory Framing

3.1 NUAA preprocessing

The function `load_nuaa_run_level()` reads the high-resolution orthogonal bundle and aggregates the raw timeseries to a run-level table using the latest timestamp and maximum wear in each grouped run. The grouping variables are `experiment_tag` and `experiment_csv_n`. The preprocessing then computes:

- t in minutes from the original timestamp in seconds,
- relative time within each experiment,
- initial wear per experiment,
- wear increment $\Delta VB = VB - VB_0$.

This choice is conceptually important. The phase-2 model does not predict raw cumulative wear from zero in a vacuum. It predicts *growth beyond the observed initial condition.* That is why initial wear is carried explicitly through the later simulator equations.

3.2 PHM 2010 preprocessing

The function `load_phm_cut_level()` reads the wear tables for cutters `c1`, `c4`, and `c6`. For each cut, flank wear is defined as the average of the three flute measurements, converted from micrometers to millimeters. The time-like variable here is not physical minute count but *relative cut index.* That is one reason PHM 2010 is used in phase-2 to validate wear shape rather than to calibrate absolute minute-based life constants.

3.3 NASA preprocessing

The function `load_nasa_case_level()` loads case-level flank wear values from `data/nasa_milling/csv/metadata.csv`. Cases are sorted by time, converted to relative time within each case, and tagged by material family:

$$\text{material 1} \mapsto \text{cast\_iron}, \qquad \text{material 2} \mapsto \text{steel}.$$

The NASA documentation confirms that wear measurements were taken at irregular intervals and that not every run has an associated flank wear observation [3]. This irregularity is compatible with the phase-2 estimation approach because the shared-exponent fit uses the observed pairs (t, VB) directly rather than assuming uniform sampling.

3.4 Exploratory implications

Even before fitting the final simulator, three exploratory observations are visible:

1. NUAA does reach wear near and slightly above the chosen transition wear of 0.25 mm, but only over a narrow range. It is best suited to early-stage trend identification, not strong late-stage inference.
2. PHM 2010 provides long wear histories in cut index, supporting the idea that the early regime is not purely linear.
3. NASA reaches far beyond the transition wear and into obvious run-to-failure territory, which makes it essential for estimating an accelerated terminal regime.

4 Modeling Strategy

4.1 Shared-exponent power law for regime identification

For each dataset role, phase-2 begins with the same model family:

$$\Delta VB_g(t) = a_g\, t^m, \tag{1}$$

where g indexes an experiment or case, a_g is a group-specific amplitude, and m is a shared exponent. This structure is fitted by nonlinear least squares in `fit_common_power()`, with one exponent shared across all groups in the same dataset slice.

The reason for this shared-exponent structure is methodological discipline. If every experiment received its own exponent, the phase-2 pipeline would lose the ability to claim an interpretable common regime. By forcing a common m, the fit asks a sharper question: *is there evidence that multiple trajectories share one regime shape, with variation absorbed mainly through amplitude?*

4.2 Candidate model check on NUAA

NUAA is the main early-regime training source, so the code compares two model families:

- linear in time: $m = 1$,
- power in time: m estimated from data.

The generated output `phase-2/outputs/nuaa_candidate_models.csv` reports:

Table 3: NUAA candidate model comparison for the early regime

Model	Exponent	RMSE (mm)
Linear in time	1.0000	0.02676
Power in time	0.6347	0.02113

The power-law model reduces NUAA RMSE by roughly 21 % relative to the linear-in-time alternative. This is the first strong empirical reason that phase-2 adopts a sub-linear early regime instead of a classical straight-line wear approximation.

4.3 Condition-intensity law from NUAA amplitudes

Once the early exponent is selected, the code estimates experiment-specific early amplitudes under a *fixed* exponent and then regresses those amplitudes against normalized process parameters:

$$\log a_g = \log k_{\mathrm{early}} + \alpha_n \log\left(\frac{n}{n_{\mathrm{ref}}}\right) + \alpha_f \log\left(\frac{f_z}{f_{z,\mathrm{ref}}}\right) + \alpha_a \log\left(\frac{a_p}{a_{p,\mathrm{ref}}}\right). \tag{2}$$

With the reference settings

$$n_{\mathrm{ref}} = 1800\,\mathrm{rpm}, \qquad f_{z,\mathrm{ref}} = 0.05\,\mathrm{mm/tooth}, \qquad a_{p,\mathrm{ref}} = 3.0\,\mathrm{mm},$$

the condition-intensity term is

$$\phi(n, f_z, a_p) = \left(\frac{n}{1800}\right)^{0.35997} \left(\frac{f_z}{0.05}\right)^{3.95601} \left(\frac{a_p}{3.0}\right)^{0.68807}. \tag{3}$$

This formulation is central to phase-2. It isolates the early wear *shape* from the process-parameter *scaling*, which makes the later simulator interpretable and analytically invertible.

4.4 Late-stage acceleration and material-family ratios

The NASA dataset is processed differently because its main value lies after the transition region. The code fixes:

$$VB_{\mathrm{tr}} = 0.25\,\mathrm{mm}, \qquad m_{\mathrm{e}} = 0.64, \qquad m_{\mathrm{l}} = 1.29.$$

For each usable NASA case:

1. identify the transition time by interpolation at $VB = VB_{\mathrm{tr}}$,
2. back out the early amplitude required to reach VB_{tr} under the selected early exponent,
3. fit a late-stage amplitude from the post-transition observations under the selected late exponent,

4. define the late-stage multiplier as
$$\rho = \frac{a_{\text{late}}}{a_{\text{early}}}.$$

This ratio-based design is prudent. It avoids claiming that the absolute NASA amplitude should replace the absolute NUAA amplitude law. Instead, NASA contributes what it can support most directly: *how much faster the late regime evolves relative to the early regime within the same case.*

4.5 Final phase-2 piecewise equation

The selected simulator law stored in `phase-2/outputs/phase2_model_summary.json` is:

$$VB(t) = VB_0 + \lambda\, k_{\text{early}}\, \phi\, t^{m_{\text{e}}}, \qquad \text{for } VB(t) \le VB_{\text{tr}}, \tag{4}$$

and

$$VB(t) = VB_{\text{tr}} + \lambda\, \rho\, k_{\text{early}}\, \phi\, (t - t_{\text{tr}})^{m_{\text{l}}}, \qquad \text{for } VB(t) > VB_{\text{tr}}, \tag{5}$$

where λ is a calibration factor for a specific tool-workpiece pair and ρ is the late-stage multiplier. The transition time is

$$t_{\text{tr}} = \left(\frac{VB_{\text{tr}} - VB_0}{\lambda\, k_{\text{early}}\, \phi} \right)^{1/m_{\text{e}}}, \qquad \text{when } VB_{\text{tr}} > VB_0. \tag{6}$$

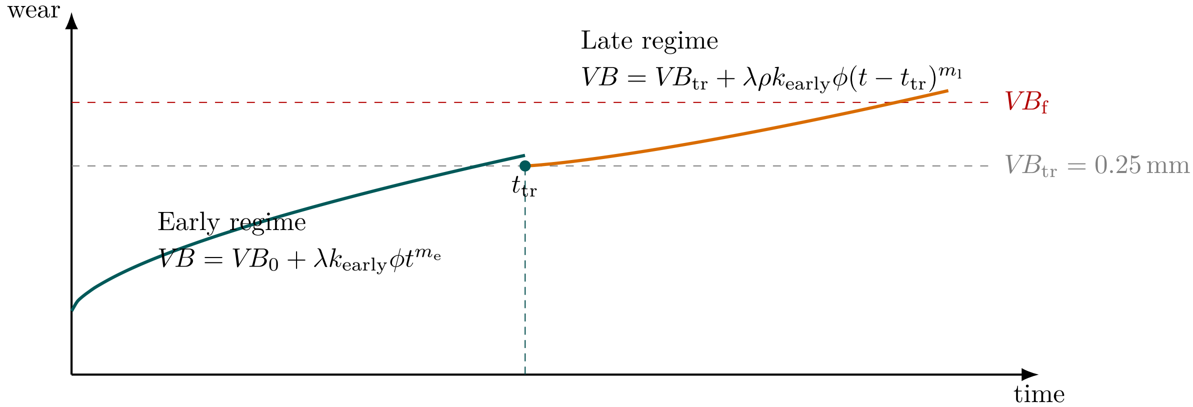

Figure 2: Native LaTeX conceptual diagram of the phase-2 piecewise wear law. The early regime is sub-linear; the late regime accelerates after the transition wear.

4.6 Analytical inversion for threshold life

The practical strength of the model is that threshold life remains available in closed form. If the target wear threshold VB_{f} lies in the early regime,

$$T_{\mathrm{f}} = \left(\frac{VB_{\mathrm{f}} - VB_0}{\lambda\, k_{\mathrm{early}}\, \phi} \right)^{1/m_{\mathrm{e}}}. \tag{7}$$

If $VB_{\mathrm{f}} > VB_{\mathrm{tr}}$, then

$$T_{\mathrm{f}} = t_{\mathrm{tr}} + \left(\frac{VB_{\mathrm{f}} - VB_{\mathrm{tr}}}{\lambda\, \rho\, k_{\mathrm{early}}\, \phi} \right)^{1/m_{\mathrm{l}}}. \tag{8}$$

That analytic invertibility is a key reason the explicit phase-2 equation remains valuable even though a simple linear regression outperforms it on one held-out NUAA benchmark, as discussed later.

4.7 Calibration from one measured point

The simulator also exposes a direct formula for λ from a measured early-stage wear point $(t_{\mathrm{obs}}, VB_{\mathrm{obs}})$:

$$\lambda = \frac{VB_{\mathrm{obs}} - VB_0}{k_{\mathrm{early}}\, \phi\, t_{\mathrm{obs}}^{m_{\mathrm{e}}}}, \tag{9}$$

provided the observation is in the early regime and exceeds the initial wear. This is scientifically useful because it separates the shared phase-2 law from specimen-specific absolute scaling.

5 Main Results

5.1 Fitted exponents and selected operating coefficients

The generated key results support a two-regime interpretation:

Table 4: Primary fitted exponents and selected coefficients

Quantity	Value	Rounded/selected	Interpretation
NUAA early exponent	0.6347	0.64	Sub-linear early wear law under orthogonal milling conditions.
PHM early exponent	0.6471	0.64	Independent confirmation that early milling wear remains near the same exponent range.
NASA late exponent	1.2873	1.29	Accelerated late-stage wear under run-to-failure behavior.
k_{early}	0.018581	0.018581	Early wear amplitude at the reference condition.
Speed exponent	0.3600	0.3600	Weakest process exponent among the three fitted NUAA factors.
Feed exponent	3.9560	3.9560	Dominant fitted sensitivity in the early condition law.
Depth exponent	0.6881	0.6881	Clear secondary effect relative to feed.
Amplitude R^2	0.7679	0.7679	Fraction of log-amplitude variance explained by the normalized NUAA parameter law.

Two observations deserve emphasis.

First, the raw exponents from NUAA and PHM are close enough that rounding to a common $m_{\text{e}} = 0.64$ is not arbitrary; it is an engineering simplification backed by two distinct early-stage datasets. Second, the feed exponent is an order of magnitude more influential than the speed exponent within the normalized early-law fit. That does not prove feed is universally the dominant milling wear driver, but it does show that within the NUAA orthogonal range, feed changes explain far more amplitude variation than spindle speed changes.

5.2 Early-regime evidence from NUAA and PHM

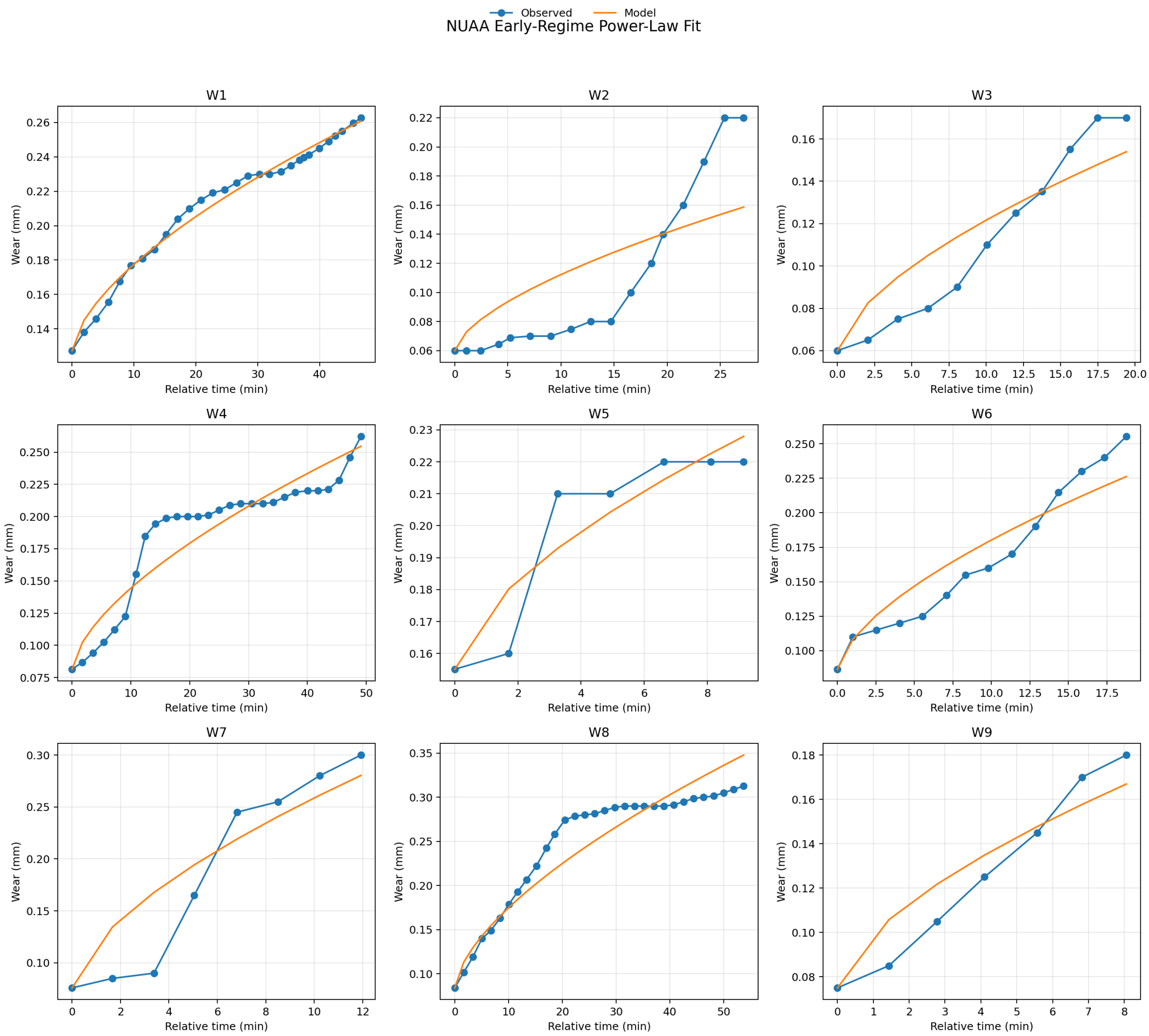

Figure 3: Observed versus modeled early-regime wear trajectories for the NUAA orthogonal experiments.

Figure 3 shows that the power-law family follows the main curvature of the orthogonal NUAA experiments without forcing a straight-line increase in wear. Experiments at higher feed and more severe combinations shift upward primarily through amplitude, which is exactly the structural assumption used later in the parameter law.

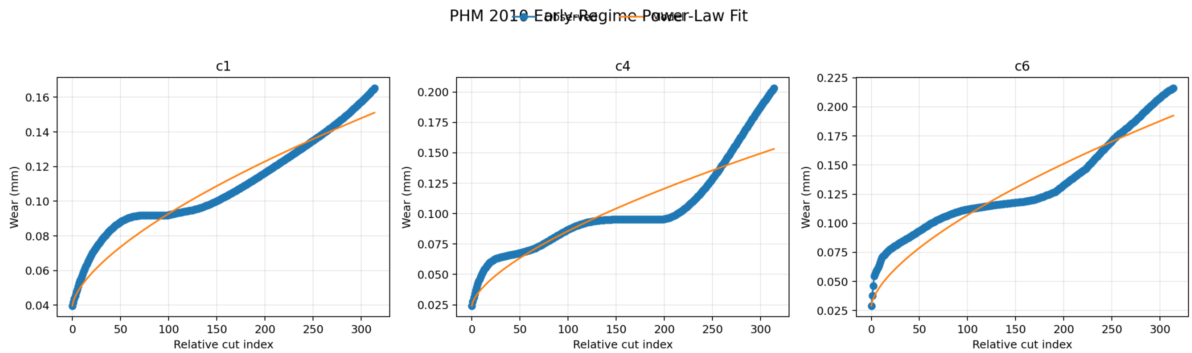

Figure 4: Observed versus modeled wear trajectories for the PHM 2010 cutter wear data, used as a cross-dataset check on early-regime shape.

Figure 4 is important because it checks whether the early-stage exponent is merely a quirk of the NUAA bundle. The fitted PHM exponent of 0.6471 says it is not. Even though the PHM time axis is cut index rather than minute count, the early wear curvature remains in the same neighborhood.

5.3 Late-regime evidence from NASA

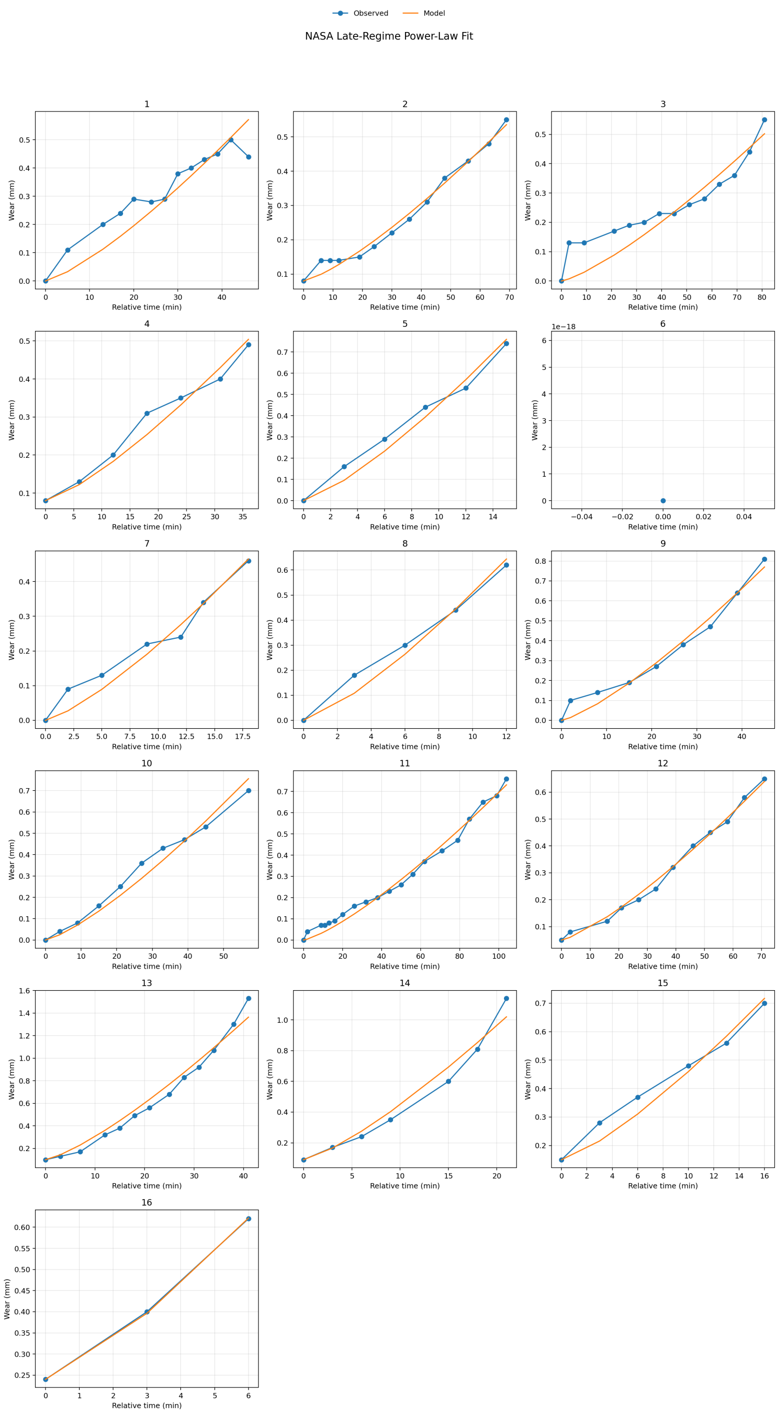

Figure 5: Observed versus modeled NASA late-regime wear trajectories. These cases provide the main evidence for end-of-life acceleration.

Figure 5 provides the empirical justification for abandoning a single-regime law. The later portions of the NASA cases are not adequately described by simply extending the early sub-linear regime. The fitted shared late exponent of 1.2873 supports an accelerating tail.

5.4 Condition-law fit quality

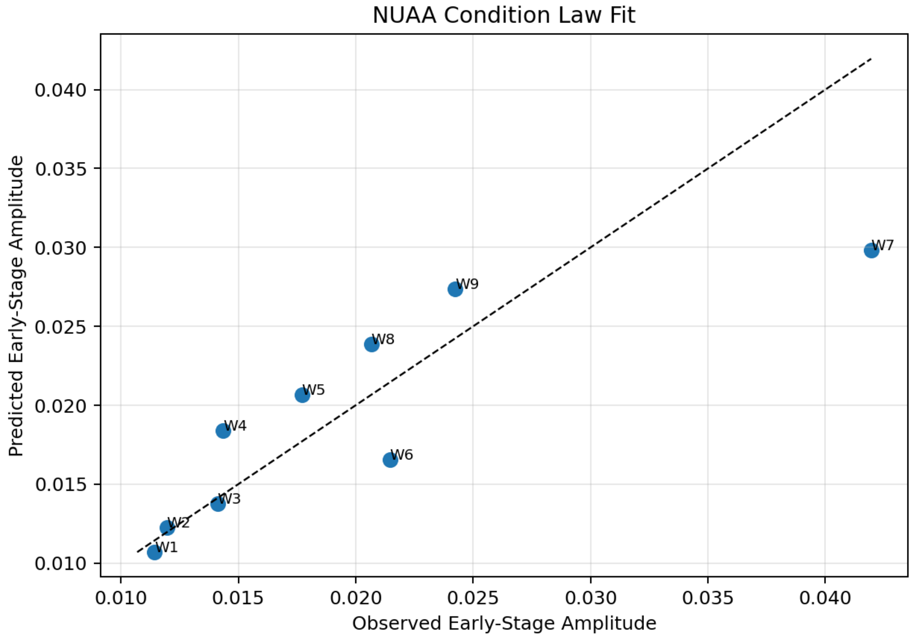

Figure 6: Observed versus predicted early-stage amplitudes under the NUAA parameter law.

The condition-law fit in Figure 6 is not perfect, but it is strong enough to justify a compact three-parameter early amplitude law. The reported amplitude $R^2 = 0.7679$ indicates that the normalized log-linear parameter relation captures most, though not all, of the amplitude variation across the nine orthogonal experiments.

5.5 Bootstrap stability

The report builder already generated bootstrap exponent distributions for the three dataset roles. The summary statistics are:

Table 5: Bootstrap summaries for the shared-exponent fits

Dataset role	Mean	Std. dev.	2.5%	Median	97.5%
NUAA early	0.7160	0.1968	0.5192	0.6450	1.2731
PHM early	0.6521	0.0519	0.5917	0.6471	0.7949
NASA late	1.2706	0.1214	1.0216	1.2752	1.4806

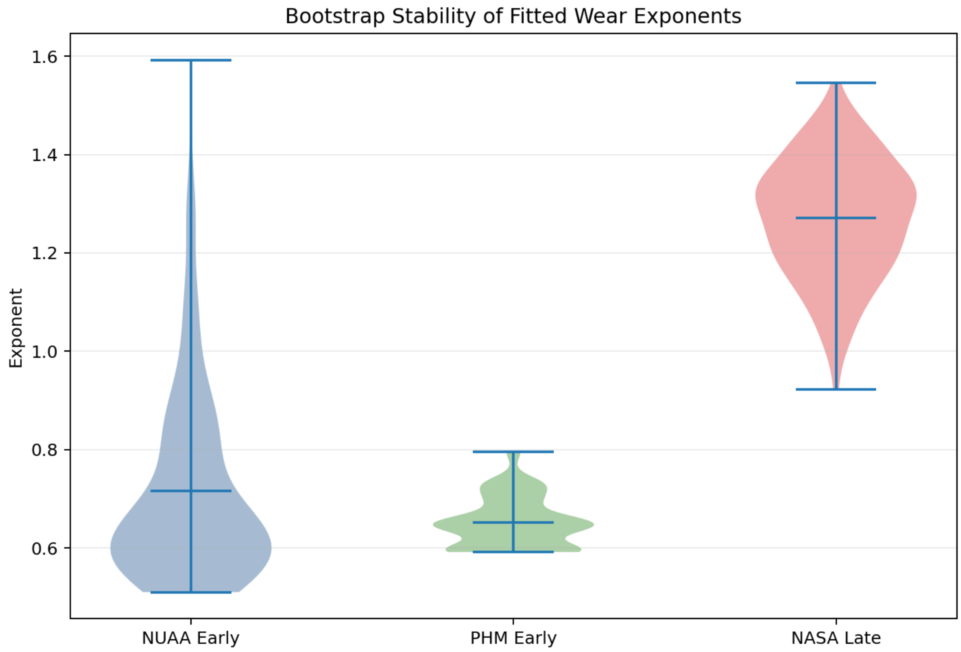

Figure 7: Bootstrap stability of the fitted exponents across the three dataset roles.

The bootstrap distributions make the research interpretation more nuanced. PHM provides the tightest support for the early exponent, centered almost exactly at the selected value of 0.64. NASA supports a clearly larger late exponent. NUAA is visibly broader. That broader distribution is not fatal, but it does warn against over-claiming numerical precision from only nine orthogonal experiments. In phase-2, the wide NUAA tail is counterbalanced by the PHM confirmation of the early-regime range.

5.6 Material-family late-stage ratios

The NASA-derived material-family summaries are:

Table 6: Late-stage multiplier summaries estimated from NASA case ratios

Material family	Count	Mean ratio	Median ratio	Suggested ρ
Cast iron	8	0.1568	0.1488	0.1488
Steel	7	0.5062	0.3994	0.3994
Generic default	–	–	–	0.2500

The interquartile ranges from the case table are approximately $[0.1352, 0.1846]$ for cast iron and $[0.2997, 0.4831]$ for steel. This means the steel late-stage ratio is not only higher in median but also substantially more variable.

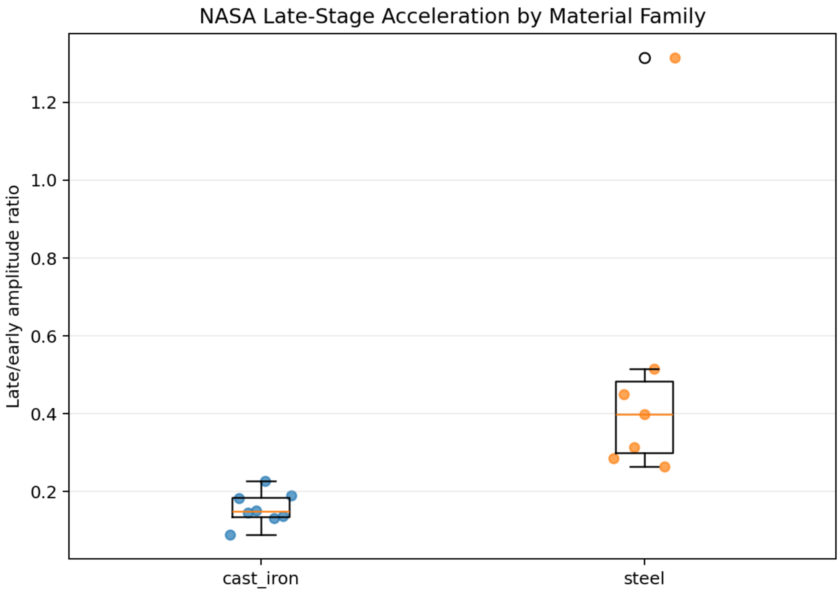

Figure 8: Material-family distribution of late-to-early amplitude ratios in the NASA cases.

One subtle but important detail is that the *generic* late-stage factor of 0.25 in the final JSON is not the median of a pooled NASA fit. It is a neutral engineering default retained by the code and then supplemented with the data-derived medians for cast iron and steel. That makes the material-specific values research-backed and the generic value pragmatic rather than strongly evidential.

6 Validation Against Alternative Predictors

6.1 Leave-one-experiment-out benchmark on NUAA

The phase-2 report builder benchmarks the explicit equation against three machine-learning baselines under leave-one-experiment-out cross-validation on NUAA:

Table 7: NUAA leave-one-experiment-out benchmark

Model	RMSE (mm)	MAE (mm)	R^2
Linear Regression	0.03358	0.02625	0.7742
Random Forest	0.03763	0.02798	0.7164
Equation	0.04371	0.03157	0.6174
XGBoost	0.04499	0.03680	0.5946

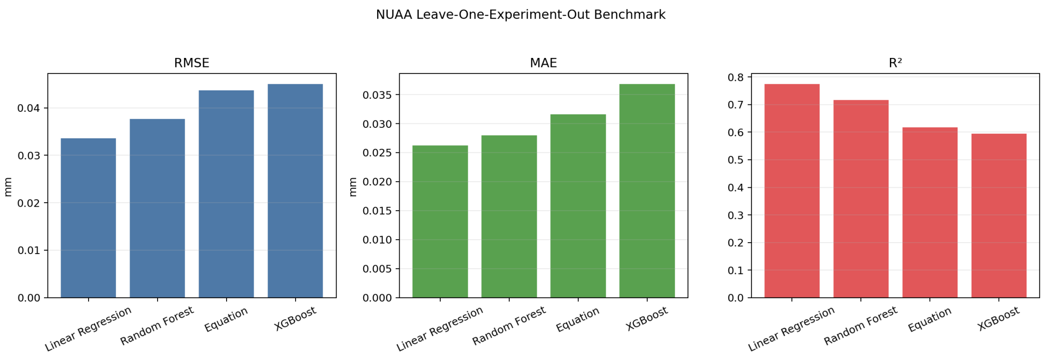

Figure 9: NUAA leave-one-experiment-out benchmark across the explicit equation and machine-learning baselines.

6.2 Interpretation of the benchmark

From a pure point-prediction perspective, the explicit equation is not the best held-out regressor on NUAA. Relative to linear regression, the equation has an RMSE disadvantage of approximately 0.0101 mm, an MAE disadvantage of 0.0053 mm, and a lower R^2 by about 0.157.

That finding should not be hidden. It is part of the honest research record. But it also does not invalidate the phase-2 equation, because the research goal is broader than pointwise curve fitting. The explicit equation retains three properties that the benchmarked ML models do not provide directly:

- closed-form inversion to target life, as in Equations 7 and 8,
- direct exposure of machining parameters in the governing law,
- explicit, interpretable separation between global regime structure and specimen-specific calibration.

In other words, the equation loses to linear regression as a held-out wear interpolator on NUAA, but it wins as a *simulator model*.

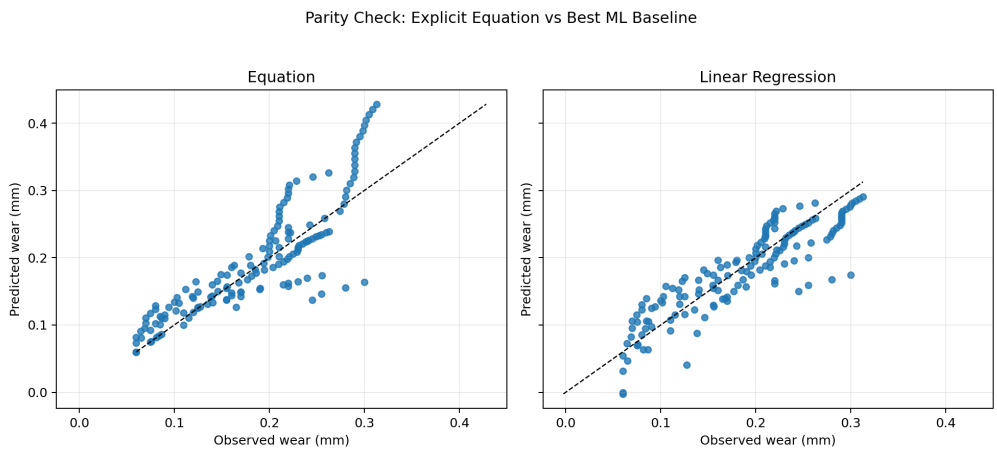

Figure 10: Parity comparison of the explicit equation and the strongest simple baseline.

Figure 10 reinforces that distinction. Linear regression tracks the held-out NUAA values more tightly, but the explicit equation remains structurally coherent and physically interpretable rather than purely statistical.

7 Simulator Behavior and Engineering Interpretation

7.1 Baseline scenario

Using `phase-2/tool_life_simulator.py` with the stored coefficients, the following baseline scenario can be evaluated:

$$\begin{aligned} n &= 1800\,\text{rpm}, \quad & f_z &= 0.05\,\text{mm/tooth}, \quad & a_p &= 3.0\,\text{mm}, \\ VB_0 &= 0.05\,\text{mm}, \quad & VB_\text{f} &= 0.30\,\text{mm}, \quad & \lambda &= 1. \end{aligned}$$

The corresponding predicted lives are:

Table 8: Illustrative simulator outputs around the phase-2 baseline

Scenario	ϕ	Life to 0.30 mm (min)	Wear at 45 min (mm)
Baseline, generic	1.0000	47.2768	0.2781
Baseline, cast iron	1.0000	50.4007	0.2667
Baseline, steel	1.0000	45.3554	0.2948
Low load (1750 rpm, 0.045 mm/tooth, 2.5 mm)	0.5756	106.7975	0.1722
High load (1850 rpm, 0.055 mm/tooth, 3.5 mm)	1.6372	23.2686	0.7596

The low-load to high-load life swing is roughly a factor of 4.59. That wide ratio is consistent with the very strong fitted feed exponent and the moderate depth effect. It is also why the simulator is useful despite the compactness of the equation: small moves in the operating point can create large moves in predicted life.

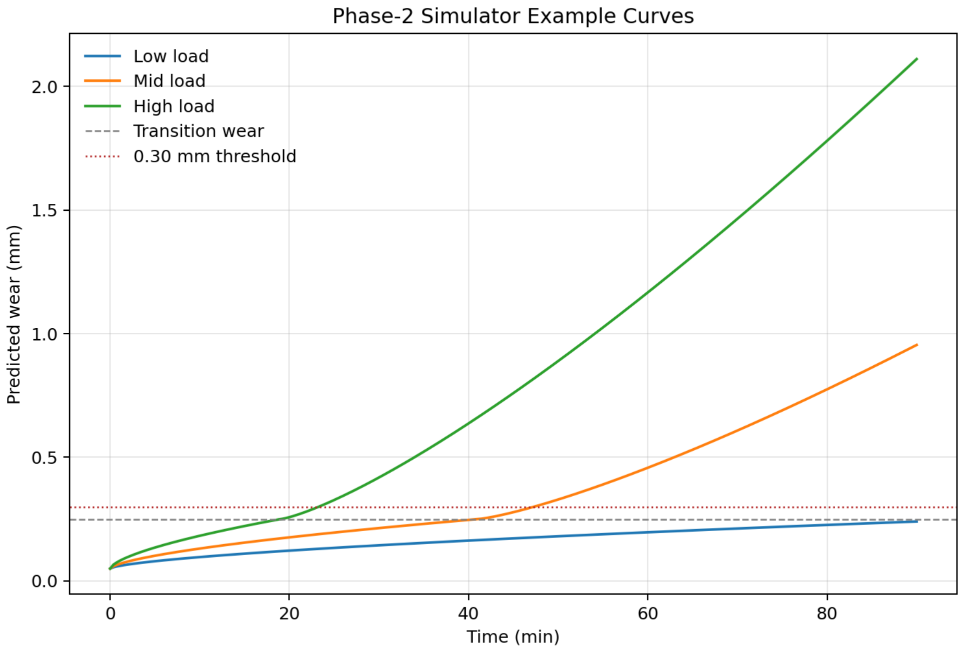

Figure 11: Example simulated wear trajectories under low, medium, and high load conditions.

7.2 Sensitivity around the reference condition

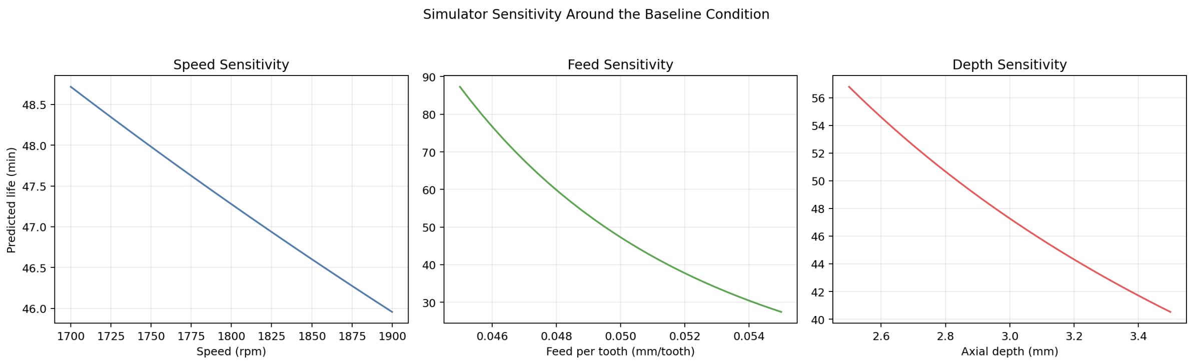

Figure 12: Predicted life sensitivity around the reference condition.

Figure 12 is especially useful for interpretation. The speed curve is comparatively shallow, which is consistent with the fitted speed exponent of only 0.3600. The feed curve is steepest, consistent with the fitted exponent of 3.9560. Depth matters, but less strongly than feed. This does not prove a universal ranking for all milling systems; it documents the ranking supported by the local phase-2 data.

7.3 Calibration workflow

The simulator is most defensible when λ is treated as a calibration quantity rather than silently fixed to one. The phase-2 codebase already includes the formula to estimate λ from one observed early wear point, and the research frontend created in the workspace exposes this interaction in the browser.

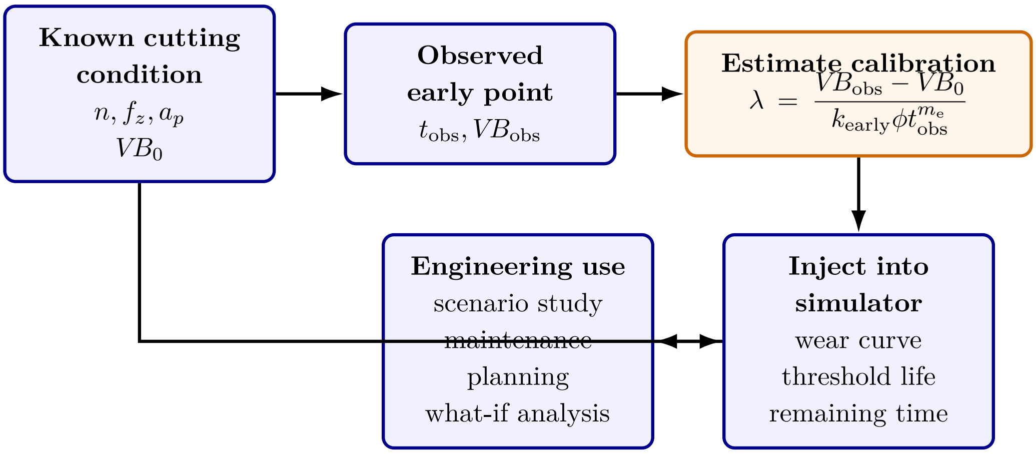

Figure 13: Native LaTeX diagram of the phase-2 calibration workflow. The explicit law becomes more credible in deployment once the global structure is calibrated to a specific tool-workpiece pair.

7.4 What the simulator should and should not be used for

From a research standpoint, the simulator is appropriate for:

- threshold-life comparison across nearby operating conditions,
- sensitivity analysis and what-if studies,
- integrating one observed wear point into a calibrated analytical forecast,
- explaining to a user how parameter changes influence predicted wear.

It is less appropriate for:

- claiming universal absolute life across unrelated tool-workpiece systems without calibration,
- extrapolating far outside the NUAA early-law parameter envelope,
- treating the generic late-stage factor as a physically grounded constant for every material.

8 Limitations and Threats to Validity

8.1 Cross-dataset stitching is both a strength and a limit

The phase-2 model is intentionally stitched across datasets. That is its key methodological innovation, but also its largest validity caveat. Early wear structure comes from one source, early-regime confirmation from another, and late-stage acceleration from a third. This is reasonable given the available public evidence, but it means the final simulator is a *research synthesis*, not a single-dataset truth model.

8.2 PHM contributes shape, not absolute minute scaling

PHM 2010 uses cut index rather than minute-based time in the wear files loaded by phase-2. Therefore PHM can confirm the presence of a similar early exponent, but it cannot by itself identify the same absolute minute-based k_{early} used in the simulator. The report should therefore not over-claim PHM as an amplitude-calibration source.

8.3 NASA contributes ratios more credibly than absolute amplitudes

NASA differs from NUAA in cutting speed representation, material system, feed definition, and failure coverage. For that reason the phase-2 code only borrows the late exponent and intra-case late/early ratios, not a wholesale transfer of absolute wear constants. This is a sound modeling decision and should remain explicit.

8.4 The generic late-stage factor is heuristic

The material-specific factors for cast iron and steel are data-derived medians. The generic value of 0.25 is not. It is a pragmatic default. This is acceptable for a simulator prototype, but future work should replace it with either

1. a pooled evidence-based estimate with uncertainty bounds, or
2. a mandatory calibration workflow that removes the need for a generic default.

8.5 The speed exponent is the least identifiable parameter

The fitted speed exponent is much smaller than the feed exponent, but the report should not interpret that as a universal physical truth. The NUAA orthogonal range for speed is narrow compared with the magnitude of the feed effect, and the report builder itself recommends acquiring a wider speed sweep before treating this coefficient as settled.

8.6 Bootstrap instability in NUAA should temper overconfidence

The broad NUAA bootstrap interval is a warning that the exact numerical early exponent is not known to three decimal places. What the evidence supports more confidently is the qualitative statement:

> the early regime is sub-linear and lies near $m \approx 0.64$, while the late regime is super-linear and lies near $m \approx 1.29$.

9 Reproducibility, Artifacts, and Folder-Level Interpretation

9.1 How the phase-2 folder fits together

The folder can be read as a compact research stack:

- `research_pipeline.py`: rebuilds the fitted coefficients, tables, and plots.
- `report_builder.py`: adds bootstrap analyses, benchmarking, additional plots, and the Word report.
- `tool_life_simulator.py`: exposes the final simulator API and CLI.
- `outputs/`: stores coefficients, summary tables, bootstrap tables, and compiled research artifacts.
- `plots/`: stores the empirical figures used by this report.
- `frontend/`: stores the static browser-based interface built on the same piecewise law.

9.2 Recommended interpretation of the deliverable

The strongest way to read phase-2 is as a *research-backed analytical simulator layer*, not as a finished production predictor. It sits between pure machine learning and purely hand-crafted Taylor-style heuristics:

- more interpretable than the ML baselines,
- more evidence-backed than an uncalibrated textbook wear law,
- still incomplete enough that calibration and continued data collection matter.

10 Conclusion

The phase-2 workspace supports a coherent research conclusion:

1. Early milling wear in the available local evidence is better modeled by a sub-linear power law than by a linear-in-time law.
2. The early wear amplitude can be related to normalized spindle speed, feed per tooth, and axial depth through a compact condition-intensity law derived from the NUAA orthogonal bundle.
3. The NASA run-to-failure cases provide strong evidence that late wear accelerates and that this acceleration depends on material family.
4. These findings can be combined into a piecewise analytical model that remains invertible for threshold life and calibratable through one observed wear point.

From a research point of view, that is the real success of phase-2. The work does not pretend to have discovered a universal milling constant. Instead, it turns heterogeneous public evidence into a simulator that is explicit about what each dataset does and does not support. That makes the model useful not only for prediction, but also for scientific critique, engineering interpretation, and future extension.

A Notation

Table 9: Notation used in the phase-2 equations

Symbol	Units	Meaning
$VB(t)$	mm	Flank wear at time t.
VB_0	mm	Initial wear state carried into the simulator.
VB_f	mm	Wear threshold used to define tool life.
VB_tr	mm	Transition wear that separates early and late regimes; fixed at 0.25 mm.
t	min	Cutting time in minutes in the simulator formulation.
n	rpm	Spindle speed.
f_z	mm/tooth	Feed per tooth.
a_p	mm	Axial depth of cut.
ϕ	–	Condition-intensity term derived from the NUAA orthogonal experiments.
λ	–	Calibration factor for a specific tool-workpiece pair.
ρ	–	Late-stage multiplier, optionally chosen by material family.
m_e	–	Early wear exponent, selected as 0.64.
m_l	–	Late wear exponent, selected as 1.29.
t_tr	min	Transition time at which the early regime reaches VB_tr.

B NUAA amplitude table used for the condition law

Table 10: Observed and predicted early-stage amplitudes from the generated `nuaa_condition_amplitudes.csv` table

Experiment	Speed (rpm)	Feed (mm/tooth)	Depth (mm)	Observed amplitude	Predicted amplitude
W1	1750	0.045	2.5	0.01143	0.01069
W2	1800	0.045	3.0	0.01196	0.01225
W3	1850	0.045	3.5	0.01411	0.01375
W4	1750	0.050	3.0	0.01436	0.01839
W5	1800	0.050	3.5	0.01771	0.02066
W6	1850	0.050	2.5	0.02146	0.01655
W7	1750	0.055	3.5	0.04195	0.02982
W8	1800	0.055	2.5	0.02066	0.02390
W9	1850	0.055	3.0	0.02424	0.02736

C Selected output inventory

- `phase-2/outputs/phase2_model_summary.json`
- `phase-2/outputs/key_results.csv`
- `phase-2/outputs/dataset_profiles.csv`

- phase-2/outputs/nasa_late_stage_case_ratios.csv
- phase-2/outputs/nasa_late_stage_ratio_summary.csv
- phase-2/outputs/nuaa_loeo_benchmark_metrics.csv
- phase-2/outputs/bootstrap_*.csv
- phase-2/plots/*.png
- phase-2/tool_life_simulator.py
- phase-2/frontend/

References

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[4] PHM Society, "2010 PHM Society Conference Data Challenge," competition overview, accessed through the source inventory recorded in `phase-2/sources.md`.

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